



Emin Yuksel

Nationality: Turkish **Date of birth:** 16/10/1997 **Gender:** Male

Phone number: (+90) 5436651154 **Email address:** eyuksel@cern.ch

Email address: eminyuksel97@gmail.com **Email address:** emin.yuksel@metu.edu.tr

Whatsapp Messenger: +905436651154

X: <https://x.com/Phyxilo>

LinkedIn: <https://www.linkedin.com/in/eminyuksel/>

GitHub: <https://github.com/Phyxilo>

ORCID: <https://orcid.org/0009-0008-7861-1879>

Home: Huzur mahallesi, 1112. cadde, 14/5 Cankaya, 06460 Ankara (Türkiye)

ABOUT ME

I am a passionate researcher and physicist. Successfully completed my undergraduate and graduate studies in the physics department of Middle East Technical University. Conducted research in the fields of Particle and Nuclear Physics. During my graduate studies, participated in the DsTau experiment at CERN. Worked on data analysis and data taking for the DsTau experiment. I have experience in numerical methods and data analysis, especially for physics problems. Due to my interest in computer sciences and being a computational physicist, have a good background in software development.

EDUCATION AND TRAINING

Bachelor of Science in Physics

Middle East Technical University [15/09/2015 – 01/02/2021]

City: Ankara | **Country:** Türkiye | **Website:** <https://phys.metu.edu.tr> | **Field(s) of study:** Natural sciences, mathematics and statistics: • Physics | **Final grade:** 2.47 | **Level in EQF:** EQF level 6 | **Type of credits:** ECTS | **Number of credits:** 281 | **Thesis:** Study of Ds Decays with Machine Learning Techniques

Link: <https://drive.google.com/file/d/1Aag9omOxkrNA61H7WO4LMH7sIPTnVE6Z/view>

Accepted into the Department of Physics at Middle East Technical University in 2015, I developed a strong foundation in both physics and computer science. My interest in game development led me to learn C, C++, and C# programming, which I applied to simulate physics concepts. Through extensive coursework in computational physics and data science, I gained expertise in data analysis, numerical methods, and particle physics, utilizing tools like Python and ROOT. My graduation project focused on separating D_s events using machine learning techniques, with a goal to contribute to the field by creating advanced computational tools.

Master's degree in Physics

Middle East Technical University [15/09/2021 – 03/09/2024]

City: Ankara | **Country:** Türkiye | **Website:** <https://phys.metu.edu.tr> | **Field(s) of study:** Natural sciences, mathematics and statistics: • Physics | **Final grade:** 3.21 | **Level in EQF:** EQF level 7 | **Type of credits:** ECTS | **Number of credits:** 234 | **Thesis:** Study of Proton Interactions in the DsTau(NA65) Experiment at the CERN-SPS

Link: <https://cds.cern.ch/record/2917039>

I continued my studies with a Master's degree in Physics at Middle East Technical University, actively participating in the DsTau experiment at CERN-SPS, where I contributed to data taking and analysis. My thesis focused on primary proton interactions in tungsten targets and a new reconstruction method I developed to select primary interactions. I presented our findings at two international physics conferences on behalf of the DsTau collaboration. I also extended my understanding of deep learning techniques and quantum computing, which I consider essential tools for data science and the future of experimental physics.

PhD degree in Physics

Middle East Technical University [15/09/2025 – Current]

City: Ankara | **Country:** Türkiye | **Website:** <https://phys.metu.edu.tr> | **Level in EQF:** EQF level 8

WORK EXPERIENCE

 **CERN** – Geneva, Switzerland

Name of unit or department: Physics - **Business or sector:** Professional, scientific and technical activities

Researcher at DsTau (NA65)

[01/02/2022 – Current]

- Since 2022, I've been involved in the DsTau/NA65 experiment at CERN.
- Played a key role in analyzing the first results of the DsTau experiment.
 - Developed a new reconstruction method and compared data with different Monte Carlo generators.
 - Participated in data-taking and emulsion development processes at CERN.

 **HyperbeeAI** – California, United States

Machine Learning & Data Engineer

[12/05/2024 – 01/11/2025]

- As a Data Engineer and Machine Learning Engineer at the startup HyperbeeAI, I have participated in numerous projects centered around large language models:
- Generated high-quality synthetic datasets to improve large language model training.
 - Scraped and curated large-scale datasets to train LLMs.
 - Trained a speech recognition model (Whisper) using Turkish-language content collected from online platforms.
 - Developed LLM-based agents and compilers to automate complex multi-step tasks.
 - Built a versatile LLM-powered smart assistant capable of managing diverse real-world workflows.
 - Designed and implemented human detection and tracking algorithms using pretrained AI models.

 **Atilim University** – Ankara, Türkiye

Name of unit or department: Physics

University Teaching Assistant

[14/10/2024 – 17/01/2025]

In 2024, I served as a part-time teaching assistant in experimental physics for engineering students at Atilim University.

 **Self** – Ankara, Türkiye

Independent AI Advisor

[01/07/2026 – Current]

Provide strategic artificial intelligence and machine learning advisory services to enterprise clients, bridging complex computational research with business operations.

Client: EGESA Holding

- Architecting the integration of a conversational AI chatbot directly into the company's enterprise resource planning (ERP) system to automate data retrieval, streamline internal communications, and optimize operational workflows.

LANGUAGE SKILLS

Mother tongue(s): Turkish

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** C1

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

C++ / C# / Python / Machine Learning / Deep Learning / ROOT / Pythia / Geant4 / Git / Windows / Linux / Microsoft Office / LaTeX / Unity / Blender / Qiskit

PUBLICATIONS

[2023]

Proceedings of Science - NA65(DsTau): Study of Tau Neutrino Production in p-A Interactions This Proceedings of Science publication, presented at the European Physical Society Conference on High Energy Physics in Hamburg, Germany, highlights the DsTau experiment at CERN-SPS. The study focuses on measuring the differential cross-section of D_s production and its decay to tau leptons in proton-nucleus interactions, utilizing nuclear emulsion detectors for high-resolution detection of decay topologies. The publication covered the physics potential of the experiment, analysis results from the pilot run, and future plans.

[2023]

Development of proton beam irradiation system for the NA65/DsTau experiment This article details the development of the target mover, a new proton irradiation system for the NA65/DsTau experiment at CERN. The target mover ensures uniform proton distribution across the detector surface, achieving a density of $1.01 \times 10^5 \text{ cm}^{-2}$ with a fluctuation of just $1.9 \pm 0.3\%$. Target mover provides optimal conditions for the reconstruction of interactions, enhancing the precision of the experiment.

[2024]

Proceedings of Science - NA65(DsTau): Study of tau neutrino production in p-A interactions This Proceedings of Science publication, presented at the 31st International Workshop on Deep Inelastic Scattering in Grenoble, France, outlines the findings from the DsTau (NA65) experiment at CERN. The research focuses on measuring the differential cross-section of D_s production and decay branching ratios in proton-nucleus interactions, using the nuclear emulsion technique for precise vertex reconstruction. The study's results, including the measurement of proton interaction length and the testing of KNO-G scaling, provide crucial data for validating Monte Carlo event generators.

[2025]

Study of Proton-Nucleus Interactions in the DsTau/NA65 Experiment at the CERN-SPS This study, published in *The European Physical Journal C* in 2025, presents the first results from the DsTau (NA65) experiment at CERN. Using nuclear emulsion detectors, the study achieved high precision in reconstructing proton interaction vertices at 400 GeV in the CERN-SPS. The findings, including the verification of KNO-G scaling in p-W interactions and the measurement of proton interaction length in tungsten, provide valuable data for improving Monte Carlo event generators and understanding proton-induced charm production.

Eur. Phys. J. C, 85 3 (2025) 230

CONFERENCES AND SEMINARS

[22/08/2023 – 29/08/2023] Hamburg

The European Physical Society Conference on High Energy Physics In 2023, attended the European Physical Society Conference on High Energy Physics in Hamburg on behalf of the DsTau experiment. During the conference, I presented the DsTau experiment's current status, shared its first results, and introduced a new reconstruction method I developed specifically for the project.

Link: <https://indico.desy.de/event/34916/contributions/146997/>

[08/04/2024 – 12/04/2024] Grenoble

31st International Workshop on Deep Inelastic Scattering Attended a conference in Grenoble, France, where I presented the first measurements of the proton interaction length in the tungsten, and also presented a detailed comparison between Monte Carlo simulations and data, along with a new reconstruction method.

Link: <https://lpsc-indico.in2p3.fr/event/3268/contributions/7332/>

[18/09/2025 – 18/09/2025] Nagoya

Advances in Proton Interaction Studies and Data-MC Comparisons in DsTau Gave a seminar in Japan on advanced analysis techniques in the DsTau experiment, covering the Proton Linking method and efficiency estimations using TrueMC. Presented results including proton interaction rate density, interaction length, and data vs. MC comparisons to identify differences and improvements to MC generators. Shared outcomes of applying the KNO-G scaling method to data and MC, and provided an overview of the RIVET framework and its applications within the experiment.

[21/06/2026 – 26/06/2026] California

Neutrino 2026: XXXII International Conference on Neutrino Physics and Astrophysics Write here thAttended Neutrino 2026 representing the DsTau (NA65) collaboration at CERN. Presented the poster "Recent results from the DsTau (NA65) experiment," which covered the first results from pilot-run data analysis, benchmarked proton vertex reconstruction against Monte Carlo predictions in high-density environments, and provided an update on the D_s meson production search.e description

PROJECTS

[31/07/2020 – 31/01/2021]

Study of Ds Decays with Machine Learning Techniques Performed particle selection using the ROOT framework and multivariate machine learning methods within the TMWA. Used kink angle and flight length variables to identify particle IDs and estimate selection efficiency.

Link: <https://doi.org/10.13140/RG.2.2.23462.48962>

[14/10/2022 – 14/11/2022]

Emulsion Development and Data Taking at CERN Collaborated with DsTau members at CERN, contributing to the chemical development of nuclear emulsion films and the irradiation of emulsion modules in the beam-line.

[25/09/2023 – 15/10/2023]

Emulsion Development at CERN Participated in the chemical development of emulsion films and contributed to their preparation for microscope scanning.

[20/08/2025 – 01/10/2025]

Emulsion Scanning at Nagoya University Performed emulsion scanning of irradiated films at CERN with the DsTau collaboration, using the new high-speed HTS2 microscope.

[01/02/2026 – 01/04/2026]

Gamma Kalkani: Scintillator-Based Gamma Detector Simulation Gamma Kalkani (Gamma Shield) Project Led the simulation and feasibility study for a scintillator-based gamma detector. Constructed detailed Monte Carlo simulations in Geant4 to accurately model radiation transport and particle interactions, enabling data-driven optimization of the detector's design prior to physical prototyping.

[14/03/2026 – 14/06/2026]

Boosted Optimization for Overcoming Subtraction Terms in NLO Generation (BOOST-NLO) Developed a machine learning pipeline using Boosted Decision Trees to eliminate negative event weights in Next-to-Leading-Order Monte Carlo simulations while perfectly preserving multi-process kinematic distributions.

Link: <https://doi.org/10.13140/RG.2.2.21398.10561>

[10/06/2026 – Current]

Gamma Kalkani: FPGA-Based Detector Prototyping and Analysis Executing the end-to-end construction of a scintillator-based gamma detector prototype. Utilizing an FPGA board as the core of the readout system to bridge physical detection events with the user interface. Actively recording initial physics measurements and analyzing the resulting datasets to calibrate the system and confirm detection efficiency against previous simulations.

CERTIFICATIONS

[06/2022 – 07/2022]

QWorld - QPrep Workshop

Link: https://drive.google.com/file/d/1LwNaVSAQ612jduQvmZ_W1CzymHgc5kJ/view?usp=drive_link

[07/2022 – 08/2022]

QWorld - QBronze (Quantum Computing and Programming)

Link: https://drive.google.com/file/d/1CUmpZcVJMfBvZsEDLHMS5ai139VEV7r/view?usp=drive_link

[08/2022 – 09/2022]

QWorld - QSilver (Quantum Computing and Programming)

Link: https://drive.google.com/file/d/1ljc3oNS0H5ap9YnMU4NtGIqgZK10dHE2/view?usp=drive_link

[09/2022 – 10/2022]

DeepLearning.AI - Neural Networks and Deep Learning

Link: <https://www.coursera.org/account/accomplishments/verify/22ZJR6AFGP62>

[10/2022 – 11/2022]

QWorld - QNickel (Quantum Computing and Programming)

Link: https://drive.google.com/file/d/1vszFNi9Cakxrj3-y0MA5HTUfRD6WUxmB/view?usp=drive_link

[06/2022 – 01/2023]

QWorld - QCOURSE 511-2

Link: https://drive.google.com/file/d/1kSxe3kvopbn38ey7raPQFoiJ_uBJ3XBS/view?usp=drive_link

[08/2023 – 09/2023]

DeepLearning.AI - Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization

Link: <https://www.coursera.org/account/accomplishments/verify/BNA95UZBU42A>

[01/2024 – 02/2024]

DeepLearning.AI - Structuring Machine Learning Projects

Link: <https://www.coursera.org/account/accomplishments/verify/B6S46LG7BRRT>

[02/2024 – 03/2024]

QWorld - QNickel (Quantum Computing and Programming)

Link: https://drive.google.com/file/d/1MPG0sT_aJoUYx1v8DF5tvddA4nxQJpsU/view?usp=drive_link

[04/2024 – 04/2024]

DIS2024 - International Workshop on Deep Inelastic Scattering

Link: https://drive.google.com/file/d/1lq9R_3j5uJt3aO7h4FpotRMO4daz7Eol/view?usp=drive_link

HOBBIES AND INTERESTS

Game Development and Physics Simulation Since 2018, I have been actively developing games with a team. We participated in Game Jams and developed numerous games and game prototypes. This process significantly contributed to the enhancement of my programming, algorithm creation, and problem-solving skills. During this time, I worked as a developer on game mechanics and also played a role in creating 3D models, shaders, VFX, and animations. Additionally, I developed a simple physics engine using the graphics language OpenGL. Overall, the simulation of physics on computers and the application of physics in game development have always fascinated me, as they are among the most effective areas for transferring physics into a digital environment.